Field and Galois Theory

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1. Field Theory

We begin by studying a fundamental invariant of fields. Recall that given any field F there is a natural ring morphism[1] $\varphi : \mathbb{Z} \to F$. Since the image of φ is an integral domain, we get that $\ker \varphi$ must be a prime ideal of $\mathbb{Z}$, that is $\ker \varphi$ is either (0) or (p) for some prime p. This leads us to our first definition;

Definition 1.1. The **characteristic** of a field F, denoted by $\operatorname{char} F$, is defined to be the nonnegative principle generator of $\ker \varphi$ where $\varphi : \mathbb{Z} \to F$ is the natural ring morphism.

In the case when $\operatorname{char} F$ is 0, notice that φ is an embedding of $\mathbb{Z}$ in the field F. Indeed, we may extend this to an embedding of its *quotient field* $\mathbb{Q}$ in F. Further, when $\operatorname{char} F$ is a prime p, then by the first isomorphism theorem, we get an embedding of $\mathbb{F}_p$ in F. It is easily checked that this is the smallest field contained[2] in F.

Infact we define the smallest field contained in F to be the *prime subfield* of F. From our discussion it is clear that the prime subfield of F must be isomorphic to either $\mathbb{Q}$ or $\mathbb{F}_p$.

Definition 1.2. If F is a subfield of a field E, then E is called an **extension** of F. We denote this by E/F (read "*E over F*").

Recall that the only ideals of a field F are the trivial ideal and itself. It follows that every ring morphism of fields is injective. Given a ring homomorphism of fields $\varphi : F \to E$, we may view E as an extension of F by identifying F with its isomorphic copy in E. ???

Moreover, the characteristic of a field is invariant under homomorphisms, hence every field is the extension of its prime subfield.

Given an extension E/F, we may view E as a F-vector space.

Definition 1.3. Let E/F be an extension. We define the **degree** of the extension as

$$[E : F] = \dim_F E.$$

Theorem 1.4. *Given a tower of fields $E/F/K$, we have the identity*

$$[E : K] = [E : F][F : K].$$

Proof. Let $[E : F] = m$ and $[F : K] = n$. Let $\{\alpha_i\}_{i=1}^m$ be a basis of E/F and $\{\beta_j\}_{j=1}^n$ be a basis of F/K. Our claim follows by showing that $\{\alpha_i\beta_j\}_{ij}$ is a basis of E/K. $\square$

[1] We require the identity to map to the identity

[2] "contained" here actually means F contains an isomorphic subfield

Let E/F be an extension. We give a few definitions.

Definition 1.5. Let $S \subseteq E$. We denote the smallest subfield of E that contains both F and S by $F(S)$. If S is finite, then $F(S)$ is said to be finitely generated.

Definition 1.6. An extension E/F is **finite** if $[E : F] < \infty$.

Definition 1.7. An element $\alpha \in E$ is **algebraic** over F if it is the root of some nonzero polynomial in $F[x]$. If α does not satisfy any such polynomial then it is called **trancendental**. The extension E/F is algebraic if all elements of E are algebraic over F.

If $\alpha \in E$ then we get a F-homomorphism $\varphi : F[x] \to F(\alpha)$ given by $x \mapsto \alpha$. Note that $\ker \varphi$ is the set of all polynomial in $F[x]$ which have α as a root. Since $F(\alpha)$ is an integral domain, $\ker \varphi$ must be a prime ideal. Further, since $F[x]$ is a PID, $\ker \varphi$ is either (0) or (p) for some irreducible $p \in F[x]$.

In the case when $\ker \varphi$ is trivial, we get that α is trancendental. This also gives that $F[x]$ is embedded in $F(\alpha)$. We may extend this to an embedding of the quotient field of $F[x]$ (denoted $F(x)$) in $F(\alpha)$. Since $F(\alpha)$ is the smallest field containing both α and F, we get $F(x) \cong F(\alpha)$.

In the case when $\ker \varphi = (p)$, then α is algebraic over F. Observe that $\operatorname{Im} \varphi \cong F[x]/(p)$ is a field (since the ideal generated by an irreducible is maximal in a PID) which contains both F and α; hence $F(\alpha) = \operatorname{Im} \varphi$. This polynomial is indeed quite special, we give it a name.

Definition 1.8. Let E/F be an extension and let $\alpha \in E$ be algebraic over F. The minimal polynomial of α in F, denoted $\operatorname{Irr}(F, \alpha)$, is the unique monic irreducible $p \in F[x]$ which has α as a root[3].

We must check the uniqueness of $\operatorname{Irr}(F, \alpha)$ before we proceed any further. From the above discussion, we get that $\ker \varphi = (p)$ for some irreducible $p \in F[x]$, since α is algebraic. Assume, without loss of generality, that p is monic. We show that $p = \operatorname{Irr}(F, \alpha)$. Suppose g is another monic irreducible with α as a root. Then $g \in \ker \varphi = (p)$, so $g \mid p$ i.e. they are associates. But since g is also monic, they must be equal.

Theorem 1.9. *If E/F is an extension and $\alpha \in E$ is algebraic over F then we have $F[x]/(\operatorname{Irr}(F, \alpha)) \cong F(\alpha)$ and $[F(\alpha) : F] = \deg(\operatorname{Irr}(F, \alpha))$.*

Proof. The first part has been proved in the discussion above. Let $p = \operatorname{Irr}(F, \alpha)$ and let $p(x) = a_0 + a_1 x^1 + \cdots + a_n x^n$ for some $a_i \in F$ with $a_n \neq 0$. We claim that $\mathcal{B} = \{1, \alpha, \ldots, \alpha^{n-1}\}$ forms a basis of $F(\alpha)/F$. Indeed they are linearly independent since any nontrivial combination would give a polynomial of smaller degree than p which has α as a root, contradicting the irreducibility of p. Now, to show $\mathcal{B}$ spans $F(\alpha)$, observe that $\alpha^n = -a_{n-1}\alpha^{n-1} - \cdots - a_0$ is in span $\mathcal{B}$. We can inductively show that $\alpha^m \in \operatorname{span} \mathcal{B}$ for all $m \in \mathbb{N}$. Since $F[\alpha] = F(\alpha)$, our result follows. □

Now we give a few theorems on finite, finitely-generated and algebraic extensions and how they relate to each other.

Theorem 1.10. *A finite extension is finitely-generated. The converse is false.*

Proof. Suppose E/F is finite. Let $\alpha_1, \ldots, \alpha_n$ be a basis of E/F. Clearly $E = F(\alpha_1, \ldots, \alpha_n)$. Conversely, consider $F(x)/F$ where x is a formal variable. The set $\{x^n\}_{n=1}^{\infty}$ is linearly independent over F, our result follows. □

Theorem 1.11. *All finite extensions are algebraic.*

[3]when viewed as a polynomial in $E[x]$

Proof. Let E/F be a finite extension, say of degree n. Take any $\alpha \in E$ then the set $\{1, \alpha, \dots, \alpha^n\}$ is linearly dependent over F. Hence, there is a nonzero polynomial that α satisfies showing that α is algebraic over F. □

Theorem 1.12. *Let E/F be an extension. If $\alpha, \beta \in E$ are algebraic over F then so is $\alpha + \beta$.*

Proof. Observe that $[F(\alpha, \beta) : F(\alpha)] \leqslant [F(\beta) : F]$ since $\operatorname{Irr}(F, \beta)$ will still have β as a root in $F(\alpha)$, hence the degree of the former extension cannot increase. By 1.4 and 1.11, $[F(\alpha, \beta) : F] \leqslant [F(\beta) : F][F(\alpha) : F]$ is finite hence algebraic. Since $\alpha+\beta \in F(\alpha, \beta)$, we get that $\alpha+\beta$ is algebraic over F. □

Corollary 1.13. *It follows that if $\alpha, \beta \in E$ are algebraic over F then $\alpha \cdot \beta$ is also algebraic. Furthermore, the algebraic elements of E hence form a field (infact, a subextension of E/F).*

Theorem 1.14. *Let $E/F/K$ be a tower of fields where E/F and F/K are algebraic extensions. Then E/K is algebraic.*

Proof. Pick any $\alpha \in E$. Then α satisfies some polynomial $p(x) = a_0 + \cdots + a_n x^n \in F[x]$. Let $S = \{a_0, \dots, a_n\} \subset F$ be the set of coefficients of p. Observe that $[K(S) : K]$ is finite since

$$[K(S) : K] \leqslant \prod_{i=0}^{n} [K(a_i) : K].$$

Moreover, α is clearly algebraic over $K(S)$. In particular, by 1.4 we have

$$[K(S, \alpha) : K] = [K(S, \alpha) : K(S)][K(S) : K] < \infty.$$

By 1.11, $K(S, \alpha)/K$ is algebraic, our claim follows. □

1.2. Algebraic Closure

Remark 1.15. Let F/K be an extension. Given elements $\alpha_1, \dots, \alpha_n \in F$, we get that

$$K[\alpha_1, \dots, \alpha_n] \subset K(\alpha_1, \dots, \alpha_n).$$

Infact, because $k[\alpha_1, \dots, \alpha_n]$ is a field. We prove this inductively.

Remark 1.16. A field F is algebraically closed if every nonconstant polynomial over F has a root. The algebraic closure of a field E is a field $\overline{E}$ such that $\overline{E}/E$ is algebraic and $\overline{E}$ is algebraically closed.

Remark 1.17. Given a field F, there is an algebraically closed field E containing F.

Proof. We construct a field E_1 in which all polynomial of $F[x]$ with degree $\geqslant 1$ have a root. For each nonconstant polynomial $f \in F[x]$, associate a symbol x_f. Let S be the set of all these symbols. Let I denote the ideal generated by $f(x_f)$ for all f in $F[S]$. We claim that I is a proper ideal. Suppose not, then there are $f_1, \dots, f_n$ (resp. $x_1, \dots, x_n$) and polynomials $g_1, \dots, g_n \in K[S]$ such that

$$g_1 f_1(x_1) + \cdots + g_n f_n(x_n) = 1.$$

Let N be such that the polynomials g_i contain only the variables $x_1, \dots, x_N$. Then our relation becomes

$$\sum_{i=1}^{n} g_i(x_1, \dots, x_N) f_i(x_i) = 1.$$

Let F' be the finite extension in which f_i have roots, say α_i for $i = 1, \dots, n$. Putting $x_i = \alpha_i$ for $i \leqslant n$ and $x_i = 0$ otherwise; we get $0 = 1$, a contradiction.

Since all ideals are contained in a maximal ideal, let $\mathfrak{m} \subset F[S]$ be a maximal ideal containing I. Then $F[S]/\mathfrak{m}$ is a field. Observe that all nonconstant $f \in F[S]$ have roots in $F[S]/\mathfrak{m}$, namely the image of x_f.

TODO □

1.3. Splitting Field and Normal Extensions

Remark 1.18. Let E/F be algebraic and $\sigma : F \to L$ be an embedding of F into an algebraically closed field L. Then there is an extension of σ to an embedding of E into L.

Remark 1.19. (Splitting Field) Let F be a field and $f \in F[x]$ of degree $\geqslant 1$. A splitting field of f is an extension E/F such that

$$f(x) = c(x - \alpha_1) \cdots (x - \alpha_n)$$

in $E[x]$ and $E = F(\alpha_1, \ldots, \alpha_n)$.

Remark 1.20. All splitting fields of $f \in F[x]$ are F-isomorphic.

Proof. Let E and K be splitting fields of $f(x) \in F[x]$. Let them have the factorisation

$$\begin{aligned} f(x) &= c_1(x - \alpha_1) \cdots (x - \alpha_n) && (\alpha_i \in K) \\ f(x) &= c_2(x - \beta_1) \cdots (x - \beta_n) && (\beta_i \in E) \end{aligned}$$

with $c_1, c_2 \in F$. Using 1.18, there is $\sigma : E \to \overline{K}$ fixing F. Then

$$f(x) = f^\sigma(x) = c_2(x - \sigma\beta_1) \cdots (x - \sigma\beta_n),$$

and due to unique factoring in $\overline{K}$, this forces $c_1 = c_2$ and $(\sigma\beta_1, \ldots, \sigma\beta_n)$ to be a permutation of $(\alpha_1, \ldots, \alpha_n)$. Hence,

$$\sigma E = F(\sigma\beta_1, \ldots, \sigma\beta_n) = F(\alpha_1, \ldots, \alpha_n) = K.$$

This proves our claim. $\square$

Remark 1.21. (Normal Extension) Let E/F be an algebraic extension. The following are equivalent and define a normal extension

1. Every embedding $\sigma : E \to \Omega$, arbitrary[4] Ω, over F is an automorphism of E. ???
2. E is the splitting field of a family of polynomials in $F[x]$.
3. The minimal polynomials of each $\alpha \in E$ splits into linear factors in $E[x]$.
4. Every irreducible polynomial of $F[x]$ which has a root in E splits into linear factors in $E[x]$.

Let $E/F/K$ be a tower. If E/F and F/K are normal, that does not necessarily imply that E/K is normal. Consider $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}$, this is not normal since $x^4 - 2$ does not have all its roots in it. We can view it as $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ and degree two extensions are normal.

1.4. Separable Extensions

Remark 1.22. (Separable degree) Let E/F be an algebraic extension. Let σ and τ be embeddings of F into algebraically closed field L and L' respectively. We assume that $L/\sigma F$ and $L'/\tau F$ are algebraic. Consider the sets S_σ of embeddings of E into L over σ, define S_τ analogously. Then there is an isomorphism $\lambda : L \to L'$ extending $\tau \circ \sigma^{-1}$ applied to σF.

$$\begin{tikzcd}
L' \arrow[dd, dash] & & L \arrow[ll, "\lambda"'] \arrow[dd, dash] \\
& E \arrow[ul, "\tau^*"] \arrow[ur, "\sigma^*"'] \arrow[d, dash] & \\
\tau F & F \arrow[l, "\tau"'] \arrow[r, "\sigma"] & \sigma F
\end{tikzcd}$$

[4] Ω is an extension of F

Observe that $\lambda_* : S_\sigma \to S_\tau$ is a bijection. Hence, the cardinality of S_σ (or S_τ) depends only on the extension E/F. Define the separable degree $[E : F]_s$ as the to be S_σ for some $\sigma : F \to L$.

(Multiplicativity of separable degree) Let $E/F/K$ be a tower. Then

$$[E : K]_s = [E : F]_s [F : K]_s.$$

If E/K is finite, then $[E : K]_s$ is finite and

$$[E : K]_s \leqslant [E : K].$$

Remark 1.23. Let E/F be an extension and $\alpha \in E$ be algebraic with $p(x) = \operatorname{Irr}(F, \alpha)$. Then α is a simple root of p if $p'(\alpha) \neq 0$.

Let E/F be an extension and let $\alpha \in E$ be algebraic over F. Then α is **separable** over F if α is a simple root of $\operatorname{Irr}(F, \alpha)$.

Let F be a field. An irreducible polynomial P over F is separable if and only if P has pairwise distinct roots in an algebraic closure of F. Observe that a polynomial $P \in F[x]$ is separable if and only if $(P, P') = (1)$.

Remark 1.24. Let F be a field. Let $P \in F[x]$ be irreducible. Then there is the dichotomy

1. P and P' are relatively prime, or
2. P' is the zero polynomial.

Case (1) occurs when $\operatorname{char} F = 0$. On the other hand, if $\operatorname{char} F = p$ then both case (1) and (2) can occur. Moreover, if case (2) occurs then $P(x) = Q(x^p)$ for some irreducible $Q \in F[x]$.

Proof. Observe that $\deg P' < \deg P$. Hence, if $P' \neq 0$, then $(P, P') = (1)$ since P is irreducible.

Assume we are in case (2) and let $P = a_0 + \cdots + a_n x^n$, then $P' = a_1 + \cdots + n a_n x^{n-1}$. If $\operatorname{char} F = 0$, then $P' = 0$ forces $P = a_1$, contradicting the irreducibility of P. If $\operatorname{char} F = p$, then $P' = 0$ forces $p \mid a_i$ whenever $p \nmid i$, i.e. $a_i = 0$. Hence $P(x) = Q(x^p)$ for some $Q \in F[x]$. Observe that Q is irreducible since a factorisation of Q would give a factorisation for P. $\square$

Remark 1.25. (Separability) Let E/F be a finite extension. The following are equivalent and define a separable extension

1. $[E : F]_s = [E : F]$.
2. Every element of E is separable over F.

Remark 1.26. Let α be algebraic over F. Then there is $\nu \geqslant 0$ such that every root of $\operatorname{Irr}(F, \alpha)$ has multiplicity p^ν and

$$[F(\alpha) : F] = p^\nu [F(\alpha) : F]_s.$$

Theorem 1.27. *(Steinitz) Let E/F be a finite field extension. E/F is simple $\iff$ there are finitely many subfields of E/F.*

Proof. Let F be finite then E is finite and $E = F(\xi)$ where ξ generates $E^\times$. We need only concern ourselves with the case where F is infinite.

TODO $\square$

Theorem 1.28. *(Primitive Element Theorem) Every finite separable extension E/F is simple.*

Proof. TODO $\square$

Remark 1.29. Let E/F be algebraic and let $\operatorname{char} F = p > 0$. The following are equivalent and define Purely Inseparable extensions.

1. $\forall \alpha \in E \setminus F$, α is not separable over F.
2. $\forall \alpha \in E$, $\exists n \geqslant 0$ such that $\alpha^{p^n} \in F$.
3. $\forall \alpha \in E$, $\operatorname{Irr}(F, \alpha) = x^{p^n} - a$ for some $n \geqslant 0$ and $a \in F$.

Remark 1.30. Let E/F be algebraic. Let $E_{sep} = \{\alpha \in E \mid \alpha \text{ separable over } F\}$. Then E_{sep} is a subfield of E/F, moreover it is the unique field $E/L/F$ such that E/L is purely inseparable and L/F is separable.

Another result is that $\operatorname{Aut}_F(E) = \operatorname{Aut}_F(E_{sep})$.

2. Galois Theory

Remark 2.1. Let $G \leqslant \operatorname{Aut}(F)$. Then the fixed field of G is

$$F^G = \{x \in F \mid x^\sigma = x \ \forall \sigma \in G\}.$$

This is a field. Observe that the automorphisms of prime subfields are trivial, hence F^G must also contain the same prime subfield as F.

Remark 2.2. An extension E/F is called **Galois** if it is normal and separable.

The group of automorphisms of an extension E/F which fixes F is denoted by $\operatorname{Aut}_F(E)$, in the case when E/F is Galois, it is denoted by $\operatorname{Gal}(E/F)$.

Remark 2.3. Let E/F be Galois and let $G = \operatorname{Gal}(E/F)$. Then $F = E^G$.

Proof. Clearly $F \subseteq E^G$, we need only prove the reverse containment. Let $\alpha \in E^G$. Since $F(\alpha)$ is algebraic over F and $F \hookrightarrow \overline{E}$, by 1.18 we get an embedding $\sigma : F(\alpha) \to \overline{E}$. Again, since E is algebraic over $F(\alpha)$, using 1.18 we can extend σ to an embedding $\sigma : E \to \overline{E}$. Since E/F is normal, σ is an automorphism of E over F, so $\sigma \in G$. Hence σ fixes α, and since σ is arbitrary we get

$$[F(\alpha) : F]_s = 1.$$

As E/F is separable, α is separable and hence $[F(\alpha) : F] = [F(\alpha) : F]_s = 1$. Therefore, $F = F(\alpha)$ and $\alpha \in F$. □

Remark 2.4. Let $E/F/K$ be a tower with E/K Galois then E/F is also Galois. Further, the map

$$F \mapsto \operatorname{Gal}(E/F)$$

from the set of intermediate fields to the subgroups of $\operatorname{Gal}(E/K)$ is injective.

Proof. It follows that E/F is separable and normal. [Lan02] p.192 [TODO]. □

Remark 2.5. Let E/F be algebraic. Let $n \geqslant 1$ be such that every element of E has degree $\leqslant n$ over F. Then $[E : F] \leqslant n$.

Proof. Let $\alpha \in E$ be such that $[F(\alpha) : F] = m$ is maximal. We contend that $F(\alpha) = E$. Suppose not, then there is $\beta \notin E \setminus F(\alpha)$. By the Primitive Element Theorem, there is $\gamma \in F(\alpha, \beta)$ such that $F(\alpha, \beta) = F(\gamma)$. But from the tower

$$F \subset F(\alpha) \subset F(\alpha, \beta),$$

we infer that $[F(\alpha, \beta) : F] > m$, hence γ has degree $> m$ over F, a contradiction. □

Remark 2.6. Let F be a field and let $G \leqslant \mathrm{Aut}(F)$ be finite. Then F/F^G is Galois and

$$[F : F^G] = |G|.$$

Proof. Let $\alpha \in F$ and let $\{\alpha_i\}_{i=1}^r$ be the orbit of α under action of G on F. Then for any $\sigma \in G$, we have $\{\sigma\alpha_i\}_{i=1}^r$ is a permutation of $\{\alpha_i\}_{i=1}^r$. Consider the polynomial

$$f(x) = \prod_{i=1}^r (x - \alpha_i),$$

observe that $f(\alpha) = 0$ and $f^\sigma = f$ for all $\sigma \in G$. This shows that all the coefficients of f are fixed by G, hence $f \in F^G[x]$. Moreover, $\mathrm{Irr}(F^G, \alpha)$ divides f, hence α is separable since the factors are distinct. Furthermore, the extension is normal since the minimal polynomial of every element splits in $F[x]$. Hence the extension F/F^G is Galois.

We now prove our second assertion. Notice that every element $\alpha \in E$ has degree $\leqslant |G|$ in F, hence by 2.5, we have $[F : F^G] \leqslant |G|$. Now, $G \leqslant \mathrm{Gal}(F/F^G)$ and $|\mathrm{Gal}(F/F^G)| \leqslant [F : F^G]_s = [F : F^G]$ since the extension is separable. Hence $[F : F^G] = |G|$ and $G = \mathrm{Gal}(F/F^G)$. □

Remark 2.7. (Galois Correspondence) Let E/F be Galois. Then

$$\{\text{Subfields of } E/F\} \longleftrightarrow \{\text{Subgroups of } \mathrm{Gal}(E/F)\}$$

Proof. Follows immediately from the last two theorems. □

Remark 2.8. Let E/F be Galois and let $\sigma : E \to \sigma E$ be an isomorphism. Then $\sigma E/\sigma F$ is Galois.

$$\begin{tikzcd} E \arrow[r, "\sigma"] \arrow[d, dash] & \sigma E \arrow[d, dash] \\ F \arrow[r, "\sigma"] & \sigma F \end{tikzcd}$$

We get that $\mathrm{Gal}(E/F)$ and $\mathrm{Gal}(\sigma E/\sigma F)$ are isomorphic (as groups). Indeed,

$$\varphi \mapsto \sigma \circ \varphi \circ \sigma^{-1}$$

maps the automorphism $\varphi \in \mathrm{Gal}(E/F)$ to an automorphism of σE over σF. This has an obvious inverse. We write

$$\mathrm{Gal}(\sigma E/\sigma F) = \sigma\, \mathrm{Gal}(E/F)\sigma^{-1} \tag{1}$$

Theorem 2.9. *Let $E/F/K$ be a tower of fields with E/K Galois. It follows trivially that E/F is Galois. Let $G = \mathrm{Gal}(E/K)$ and $H = \mathrm{Gal}(E/F)$.*

The extension F/K is Galois if and only if $H \trianglelefteq G$. In which case, $\mathrm{Gal}(F/K) \cong G/H$.

Proof. ($\Rightarrow$) Let F/K be Galois and let $G' = \mathrm{Gal}(F/K)$. Consider the mapping $\pi : G \to G'$ where $\sigma \mapsto \sigma\,|_F$. This is a group homomorphism since F/K is normal and hence any embedding of F is an automorphism of F by 1.21. This map is surjective since any element $\tau \in G'$ extends to a embedding of E in $\overline{E}$ which is an automorphism of E since E/F is normal. Indeed, $\ker\pi = H$ hence $G' \cong G/H$.

($\Leftarrow$) Assume F/K is not Galois, then it is not normal since separability is forced [ref]. TODO [Lan02] p.196. □

Remark 2.10. Let K/F be Galois with group G and E/F an arbitrary extension. Assume that K and E are subfields of some field. Then KE/E is Galois and $K/K \cap E$ is Galois with Galois groups H and G' respectively. Moreover, $H \cong G'$.

$$\begin{tikzcd}
& KE \arrow[dl, dash] \arrow[dr, dash, "H"] & \\
K \arrow[dr, dash, "G'"] \arrow[ddr, dashed, dash, "G"'] & & E \arrow[dl, dash] \\
& K \cap E \arrow[d, dash] & \\
& F &
\end{tikzcd}$$

Proof. Let $\sigma \in H$, then $\sigma\mid_K \colon K \to KE$ is an automorphism of K by 1.18. Consider the map $\Psi : H \to G$ given by $\sigma \mapsto \sigma\mid_K$, this is clearly a homomorphism. We shall prove that this is the required isomorphism of H and G' (we don't know the image of Ψ *apriori*, so the codomain is G for safety). This map is injective, indeed if $\sigma\mid_K = id_K$ then $\sigma = id_{EK}$ since EK is made of sums and products of E and K and σ already fixes E. Let $H' = \operatorname{Im}\Psi$, then H' fixes $E \cap K$. Conversely, if $\alpha \in K$ is fixed by H' then α is fixed by H so $\alpha \in E$. So $K^{H'} = K \cap E$. By 2.6, it follows that $H \cong G'$. □

Theorem 2.11. *lang thing*

Remark 2.12. Let E/F be an algebraic extensions with $G = \operatorname{Aut}_F(E)$. Let $f \in F[x]$ with $f \neq 0$ and $\Omega = \{\alpha \in E \mid f(\alpha) = 0\}$. Assuming Ω is nonempty, the following hold:

(a) The action of G permutes the elements of Ω.

(b) If $E = F(\Omega)$, then G is embedded into S_Ω.

(c) If f is irreducible and E/F if Galois then G acts transitively on Ω.

2.1. Examples

Remark 2.13. Example: E/F is of degree two $\iff$ E is the splitting field of $x^2 - c$ where $c \in F$ is not a square.

Proof. If E/F is of degree two then $E = F(\alpha)$ for any $\alpha \in E \setminus F$, indeed $[E : F(\alpha)][F(\alpha) : F] = 2$. Then minpoly of α, call it f, has degree two, say $f(x) = x^2 + 2\lambda x + \mu$. Now,

$$0 = \alpha^2 + 2\lambda\alpha + \mu = (\alpha + \lambda)^2 + (\mu - \lambda^2)$$

Replacing α by $\alpha + \lambda$, the minpoly now becomes $f(x) = x^2 - c$ for some $c \in F$ which is not a square. The converse is obvious.

If $\operatorname{char} F \neq 2$, the f is separable and hence E/F is Galois, say $G = \operatorname{Gal}(E/F)$. Using 2.6, $|G| = 2$ and hence $G \cong \mathbb{Z}/2\mathbb{Z}$. More concretely, let σ be the non-identity element of G, then $\sigma\alpha = -\alpha$. □

3. Unsorted

3.1. The Frobenius Endomorphism

A field F is **perfect** if every algebraic extension is separable.

Definition 3.1. (Frobenius endomorphism) Let F be a field of prime characteristic p. Define the endomorphism $\varphi : F \to F$ given by $x \mapsto x^p$. This is indeed an endomorphism since $p \mid \binom{p}{r}$ for $0 < r < p$. The image of φ is denoted by F^p.

Note that this is not necessarily surjective. Consider $\mathbb{F}_p(t)$ where t is trancendental. Then the image of φ does not contain t. If it did, then there would be $f(t), g(t) \in \mathbb{F}_p[t]$ such that $f^p(t)/g^p(t) = t$. But the degree of $f^p(t)/g^p(t)$ is $p(\deg f - \deg g)$. Since p divides the degree, it cannot be 1, which is the degree of t.

Theorem 3.2. *The field F is perfect if and only if either $\operatorname{char} F = 0$ or $\operatorname{char} F = p$ and $F^p = F$.*

Proof. Let E/F be algebraic and let $\alpha \in E$ with $P(x) = \operatorname{Irr}(F, \alpha)$. If $\operatorname{char} F = 0$, then $(P, P') = (1)$ showing that P has no repeated roots, so α is separable.

If $\operatorname{char} F = p$ and $F^p = F$. If $(P, P') = (1)$, then our result follows by repeating the previous argument. If $P' = 0$, then $P(x) = Q(x^p)$ for some irreducible $Q(x) = a_0 + \cdots + a_n x^n \in F[x]$. Since the Frobenius map is an isomorphism, we get $b_i \in F$ such that $b_i^p = a_i$. Let $\widetilde{Q}(x) = b_0 + \cdots + b_n x^n$. It follows that $\widetilde{Q}^p(x) = P(x)$, contradicting the irreducibility of P.

Conversely, suppose $\operatorname{char} F = p$ and F is perfect. For any $\alpha \in F$, consider $x^p - \alpha \in F[x]$. If this polynomial does not have a root in F,

TODO $\square$

It follows that all finite fields are perfect.

3.2. Roots of Unity

Let F be a field. For an integer $n \geqslant 1$, the n^{th} root of unity in F is the set of roots of the polynomial $x^n - 1$. We denote this set by $\mu(n)$. Clearly $|\mu(n)| \leqslant n$ since $x^n - 1$ has atmost n roots. Moreover, $\mu(n)$ is a group under multiplication and is cyclic since any finite subgroup of $F^\times$ is cyclic. The generator of the group $\mu(n)$ is said to be a *primitive* n^{th} root of unity.

TODO

Remark 3.3. The n^{th} cyclotomic polynomial is the product over all primitive n^{th} root, that is

$$\Phi_n(x) = \prod_{(k,n)=1} (x - e^{2\pi i k/n}) \in \mathbb{C}[x].$$

We list a few results about them.

1. $\deg \Phi_n(x) = \varphi(n)$, where φ is the euler totient function.
2. $x^n - 1 = \prod_{d|n} \Phi_d(x)$.
3. $\Phi_n(x) \in \mathbb{Z}[x]$ monic.
4. $\Phi_p(x) = x^{p-1} + x^{p-2} + \cdots + x + 1$.
5. $\Phi_n(x)$ is irreducible in $\mathbb{Q}[x]$, and hence in $\mathbb{Z}[x]$ by Gauss's Lemma.
6. $\Phi_n(x) = \operatorname{Irr}(\mathbb{Z}, \zeta_n)$.

Define the n^{th} cyclotomic field $\mathbb{Q}_n = \mathbb{Q}(\zeta_n)$. A few facts are listed below.

1. It is the splitting field of $x^n - 1$.
2. $\mathbb{Q}_n/\mathbb{Q}$ is Galois.

3. $\mathrm{Gal}(\mathbb{Q}_n/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$
4. $\mathbb{Q}_n \cong \mathbb{Q}[x]/(\Phi_n(x))$.
5. $[\mathbb{Q}_n : \mathbb{Q}] = \varphi(n)$.

3.3. Solvability

TODO

3.4. Abelian Extensions

Theorem 3.4. *(Kummer Extensions) Let $F \subset \mathbb{C}$ contain ζ_p and let E/F be finite. E/F is Galois and $[E : F] = p \iff E = F(\alpha)$ where $\alpha \in E \setminus F$ and $\alpha^p \in E$.*

Proof. ($\Leftarrow$) Let $a = \alpha^p \in F$ and let $f(x) = x^p - a \in F[x]$. Observe that f is irreducible (??). The splitting field of f will be $F(\alpha, \zeta_p) = E$. Since f is irreducible, it is separable (char 0), hence E/F is Galois with $[E : F] = p$.

($\Rightarrow$) TODO □

Theorem 3.5. *(Artin-Schreier Extensions) Let F be a field of characteristic $p > 0$. Let $f \in F[x]$ be such that $f(x) = x^p - x - c$. Then we have the dichotomy;*

- *f has a root in F, in which case all roots of f are in F.*
- *f is irreducible, in this case $\mathrm{Gal}(f) \cong \mathbb{Z}/p\mathbb{Z}$*

Proof. Let $\alpha \in \overline{F}$ be a root of f. Observe that for $0 \leqslant n < p$,

$$\begin{aligned} f(\alpha + n) &= (\alpha + n)^p - (\alpha + n) - c \\ &= f(\alpha) + (n^p - n) = 0. \end{aligned}$$

Hence the set of roots of f will be $\{\alpha + n \mid 0 \leqslant n < p\} \subset \overline{F}$.

Suppose $\alpha \notin F$, let $g = \mathrm{Irr}(F, \alpha)$. Then $g \mid f$ TODO □

References

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